



Round 2 Section 3 – Case Study Information Pack

Section 3: Case Study – Breaking Mews

Relates to Questions 24-38

30 Marks available in this Section

INTRODUCTION

You are working for an insurance company and your company is looking at introducing a new product that covers cats. They would like your help to determine the prices/profitability of the product specifically by using an example customer with three cats. The analysis you are doing starts from 1 January 2019.

The three cats are Bruce, Earle and Richard and are aged 0, 8 and 15 respectively. They are looking at getting cover of \$1,000, \$750 for Bruce and Earle respectively. For Richard they want to get the amount of cover that gives the same premium as for Earle. For this analysis all three cats should be looked at in isolation of each other.

EXPECTED CAT MORTALITY PROBABILITIES

The probability that a cat currently aged 0 dies at each age is included in the associated spreadsheet. The first three rows of this data are:

Age at beginning of year	Probability of cat currently 0 dying at age
0	0.0550
1	0.0400
2	0.0260

The following explanations may be useful in understanding the mortality probabilities:

- The probability of a cat currently aged 0 dying when 1 year old is 0.0400 (4%)
- The probability of a cat currently aged 0 dying when 2 years old or less (before the age of 3) is 0.1210 (12.1%)
- The total of all the probabilities added together is 100% (as it is certain the cat must die at some point).
- You may wish to use the data to calculate the probability that a cat, having reached a certain age, will then die in the next year.

For this exercise assume that cats have a birthday at 31 December each year.

EXPECTED PREMIUMS

The product has a premium to be paid annually in advance (on 1 January each year). The premium is the same amount each year.

After the first premium is paid, each year there is a chance that a customer will not pay their premium which will result in them forfeiting the policy (and it can not be reinstated). This probability is 3% per annum for a cat aged 10 or younger on 1 January, while it is 1% for those 11 or older.

EXPECTED CLAIMS

If the pet dies in that calendar year a payment will be made on 31 December. The claim amount paid on death is chosen by the owner of the cats with the premium priced accordingly.

Due to the potential for owners getting cat insurance when they know their cat is about to die, if a cat dies in the first year of the policy then the claim amount is only equal to the total amount of any premium paid.



Round 2 Section 3 – Case Study Information Pack

EXPECTED ADMIN FEES

Your company believes that each policy will cost \$2.50 per policy in 2019 to administer which will increase with inflation of 2% per annum. This is paid to an administrative company on 1 January each year. This amount is paid each year the customer pays their premium.

EXPECTED WRAP-UP FEES

There is a policy wrap up fee of \$20. This is paid either on 1 January the year after the cat dies, or 1 January the year in which a policy is forfeited. This amount does not increase with inflation.

REGULATORY CAPITAL BALANCE

The insurance legislation in the country mandates an amount of regulatory capital must be held against each policy to cover the expected claims payments.

At 1 January this is required to be the larger of:

1. Discounted future expected claims payments – discounted future expected premiums
2. $(\text{Expected claims payments on 31 December of that year}) / 1.03$

For the calculation in 1 above, cashflows should be discounted to 1 January of that year at 10% per annum. Note that the premium already paid on 1 January is excluded from the required capital calculation.

Interest is earned at 31 December of each year. This is calculated at 3% per annum of the regulatory capital balance as at 1 January of the same year.

EXPECTED TRANSFERS TO/FROM THE WIDER COMPANY

Each year, cash must be transferred between a specific policy and the wider company to ensure that the closing cash for a specific policy is equal to the regulatory capital balance required for that policy (no more and no less).

This is effectively the net cash flow that the company pays and receives from a specific policy, and how they generate a profit from each policy.

The premium is set such that the profile of the expected transfers generates a specified return for the policy (explained further below).

OPTIMAL PREMIUM CALCULATION

The company wants you to set the optimal annual premium such that discounting the expected transfers (as described above) at 5% per annum to 1 January 2019 is equal to 0.



Round 2 Section 3 – Case Study Information Pack

EXAMPLE

On 1 January of each year you need to calculate the required regulatory capital and the associated cash movements to achieve it for each policy. The following example for 1 January 2020 may help in calculating the capital balance as at 1 January of each year.

Description	Notes
1 January 2019 closing balance	Equals the closing regulatory capital balance as at 1 January 2019
Plus: Interest on regulatory capital balance	Interest earned on the cash balance in 2019
Minus: Expected claims	Expected claims paid on 31 December 2019
Equals: 31 December 2019 closing balance	
Plus: Expected premiums	Expected premiums earned on 1 January 2020
Minus: Expected admin fees	Expected admin fees paid on 1 January 2020
Minus: Expected wrap-up fees	Expected wrap-up fees paid on 1 January 2020
Plus/Minus: Transfer to/from wider company	Calculated such that the 1 January 2020 closing balance equals the required regulatory capital.
Equals: 1 January 2020 closing balance (closing regulatory capital balance)	The regulated cash balance that must be maintained, at 1 January 2020

OTHER NOTES

All results should be calculated on an expected value basis, ie premiums are multiplied by the probability the premium is paid (ie the cat is alive and the policy has not been forfeited) to determine the expected premium in a year.

For discounting purposes 31 December is treated as the same day as 1 January of the following year.

Do not round interim calculations for any of these questions.

You may wish to use the questions to help guide you on the interim calculations required to build the model.

Prepare your model and then use it to answer questions 24-37.

When finished, please upload your workbook (Question 38).



Round 2 Section 3 – Case Study Information Pack

QUESTIONS

Question 24

What is the probability of a cat currently aged 0 living to the age of 9? [1 mark]

- A. 75.1%
- B. 75.2%
- C. 75.3%
- D. 75.4%
- E. 75.5%
- F. 75.6%
- G. 75.7%
- H. 75.8%
- I. 75.9%

Question 25

What is the probability of a cat currently aged 6 dying in the next year? [1 mark]

- A. 2.32%
- B. 2.33%
- C. 2.34%
- D. 2.35%
- E. 2.36%
- F. 2.37%
- G. 2.38%
- H. 2.39%
- I. 2.40%



Round 2 Section 3 – Case Study Information Pack

Question 26

What is the probability that Bruce's owner will pay a premium for him on 1 January 2021?
[2 marks]

- A. 85.07%
- B. 85.08%
- C. 85.09%
- D. 85.10%
- E. 85.11%
- F. 85.12%
- G. 85.13%
- H. 85.14%
- I. 85.15%

Question 27

What is the probability that a claim will be made for Bruce dying in the 2025 calendar year?
[2 marks]

- A. 1.57%
- B. 1.58%
- C. 1.59%
- D. 1.60%
- E. 1.61%
- F. 1.62%
- G. 1.63%
- H. 1.64%
- I. 1.65%



Round 2 Section 3 – Case Study Information Pack

Question 28

If the premium was set at \$50 for Bruce, what is the total expected claims payment to Bruce across all years? [2 marks]

- A. 700.44
- B. 700.45
- C. 700.46
- D. 700.47
- E. 700.48
- F. 700.49
- G. 700.50
- H. 700.51
- I. 700.52

Question 29

In a policy for Bruce, what are the total expected admin expenses paid over the life of the policy? [2 marks]

- A. 31.12
- B. 31.13
- C. 31.14
- D. 31.15
- E. 31.16
- F. 31.17
- G. 31.18
- H. 31.19
- I. 31.20



Round 2 Section 3 – Case Study Information Pack

Question 30

If the premium was set at \$50 for Bruce, what is the expected regulatory capital required at 1 January 2024? [2 marks]

- A. 68.82
- B. 68.83
- C. 68.84
- D. 68.85
- E. 68.86
- F. 68.87
- G. 68.88
- H. 68.89
- I. 68.90

Question 31

Leaving the premium at \$50 for Bruce, what is the expected transfer to the wider company on 1 January 2019? [2 marks]

- A. 44.81
- B. 44.82
- C. 44.83
- D. 44.84
- E. 44.85
- F. 44.86
- G. 44.87
- H. 44.88
- I. 44.89



Round 2 Section 3 – Case Study Information Pack

Question 32

Leaving the premium at \$50 for Bruce, what is the expected transfer from the wider company on 1 January 2030? [2 marks]

- A. 15.72
- B. 15.73
- C. 15.74
- D. 15.75
- E. 15.76
- F. 15.77
- G. 15.78
- H. 15.79
- I. 15.80

Question 33

What is the optimal premium for the company to charge for Bruce annually? [3 marks]

- A. 55.10
- B. 55.11
- C. 55.12
- D. 55.13
- E. 55.14
- F. 55.15
- G. 55.16
- H. 55.17
- I. 55.18



Round 2 Section 3 – Case Study Information Pack

Question 34

What is the optimal amount of premium to charge for Earle annually? [3 marks]

- A. 82.27
- B. 82.28
- C. 82.29
- D. 82.30
- E. 82.31
- F. 82.32
- G. 82.33
- H. 82.34
- I. 82.35

Question 35

What amount of cover is selected for Richard? Enter the amount to the nearest cent. [3 marks]

Question 36

Bruce's owner has talked to a competitor who has the same product in the market, only the premium is \$50. Assume the competitor has the same assumptions other than the rate they use to discount transfers to/from the wider company. What would the competitor charge as an annual premium for Earle? Enter the amount to the nearest cent. [2 marks]

Question 37

For this question, the regulator is looking at changing the investment requirements for regulatory capital. Instead of all regulatory capital earning a 3% pa interest rate, the regulator will allow the amounts over the 2nd criteria (ie (expected claim payments that year)/1.03) to earn a 10% pa interest rate. What is the new optimal premium for Bruce? [3 marks]

Question 38

Please upload your workbook for this section.